

1. Problem Definition & Motivation

Microplastics, defined as plastic particles measuring 5 mm or smaller, represent one of the most pervasive and poorly understood pollutants in aquatic environments. They originate from the fragmentation of larger plastic debris and synthetic materials, and have been widely detected in oceans, rivers, reservoirs, and even drinking water supplies worldwide. Unlike large floating plastics that can be visually detected, <5 mm particles are invisible to traditional monitoring approaches and they can disperse across vast water bodies driven by currents and wind.

The core problem this project addresses is the absence of scalable, real-time, on site monitoring infrastructure for microplastic concentration and spatial distribution. Current detection pipelines require water samples to be retrieved physically and analyzed in laboratory settings using Fourier transform infrared spectroscopy (FTIR), Raman spectroscopy, or optical microscopy. Each of these are expensive, slow, and rare to find in areas where this problem is greatest. We are then left with 3 fundamental blind spots: we cannot track where microplastics are accumulating, how concentration gradients evolve seasonally, or whether remediation efforts are effective.

This project proposes to close this gap by designing and evaluating a system in which a lightweight neural network, optimized for edge deployment, runs directly on an IoT device (e.g., Raspberry Pi), theoretically mounted underneath a buoy. The device continuously samples water imagery using a microscope camera combined with controlled multi-wavelength LED illumination, detects and counts microplastic particles, and transmits aggregated density metadata for geospatial tracking.

Autonomous, low-cost sensors distributed across waterways could enable the first case of high-resolution temporal maps of microplastic dispersion, which can improve environmental policy and cleanup logistics.

2. State-of-the-Art (SOTA) Analysis

Laboratory spectroscopy methods

FTIR and Raman spectroscopy are the standard for polymer identification, achieving near-perfect classification accuracy. Machine learning, specifically SVMs and CNNs are applied to spectral signatures, which has accelerated throughput significantly. However, these systems require specialized hardware, sample preparation, and trained operators, making them incompatible with autonomous deployment in the field. Another physical

limitation we have is that standard spectroscopic methods cannot reliably resolve particles below 10–20 μm .

Vision-based deep learning systems

Recent work has applied YOLO-type detectors to optical microscopy images of microplastics with promising results. The Plastiview system demonstrated that a Raspberry Pi 4B paired with a USB digital microscope and a YOLOv5s model could achieve a mAP@50 of 0.748 in controlled settings. Plastiview researchers also showed real-time detection of microplastics in motion using camera-based AI, measuring particle size and velocity via DeepSORT tracking. An IoT-integrated system pairing YOLOv8 with a microscopic camera achieved multi-class classification of microscopic polymer types with a cloud-based dashboard for visualization (which would also be my solution to integrating temporal data to create maps).

IoT-enabled portable spectrometers

Researchers also developed an IoT-enabled spectrometer achieving 96% detection accuracy for microplastics in food samples. Other researchers have also designed a low-cost Raman spectrometer with IoT features capable of detecting microplastics in water at very low concentrations. Emerging research uses compact optical sensors with AI-enabled cameras deployed on drones and buoys for near real-time environmental tracking.

Gaps this project targets

Despite this progress, critical gaps remain:

1. No existing system evaluates the tradeoff between model compression techniques (quantization, pruning) and detection accuracy specifically in the context of aquatic microplastic monitoring
2. Most vision-based systems are evaluated under controlled lab conditions and fail to account for turbidity, biofouling, and organic false-positive contamination in real water environments

3. Proposed Solution & Technical Challenges

System architecture

The proposed BuoyNet system consists of three integrated components:

1. A sensing module comprising a Raspberry Pi 4B or Pi Zero 2W, a USB digital microscope with controlled multi-wavelength LED illumination
2. An inference module running a quantized lightweight CNN (MobileNetV3-Small or YOLOv8-nano) for microplastic detection and classification
3. A communication module transmitting aggregated particle count, average size estimate, and timestamped readings via WiFi at intervals.

The way I will test this is by creating the BuoyNet model and communication module software, then testing whether the energy, latency, and memory requirements fall in line with the hardware I proposed.

Model design and compression

The core neural network will be trained on the MPID dataset, labeled microplastic image dataset and fine-tuned for the specific imaging conditions of the underwater camera setup. Model compression is the central deep learning objective: applying INT8 post-training quantization using PyTorch Mobile, evaluating structured pruning, and measuring the accuracy-latency-power Pareto frontier on the target hardware. I will also conduct resolution ablations as prior IoT research has shown that reducing input resolution from 224x224 to 128x128 can halve inference cost while maintaining maximum accuracy.

Feasibility

The hardware pipeline is validated by Plastiview and similar prior systems. Public microplastic image datasets exist for training. PyTorch Mobile and ONNX Runtime for ARM are mature frameworks for Pi deployment. The main risk is the gap between lab datasets and real aquatic imagery which I will address through data augmentation in order to simulate turbidity and lighting. A second risk is that the smallest microplastics (<1 mm at camera distance) could fall below the optical resolution of low-cost microscopes, raising my size floor for detection.

Challenges and mitigations:

- Simulating turbidity and visual noise: Apply image preprocessing (adaptive histogram equalization, background subtraction) and train with augmented turbidity overlays
- Class imbalance in microplastic datasets: Use focal loss and oversampling of rare polymer classes.
- Inference latency on Pi hardware: Profile with INT8; target <500ms per frame to support duty-cycled sampling at 2 frames/minute.
- False positives from organic matter: Incorporate morphological filters (circularity, aspect ratio thresholding) as post-processing to reduce non-plastic debris misclassification.

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